

ARMSTRONG



DEMATIC

EFS TestCaseGenerator

User Manual

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Table of Contents

- 1 Introduction**
- 2 Login**
- 3 Project Dashboard**
- 4 Project Cards**
- 5 Working Inside a Project**
- 6 Master Dashboard**
- 7 Edit Test Case Library**
- 8 Best Practices**
- 9 Frequently Asked Questions**
- 10 Glossary**

1. Introduction

1.1 Overview

The EFS Test Case Generator is a comprehensive, standalone desktop application designed specifically to streamline and standardize the testing and commissioning of automated warehouse equipment. It provides a secure, organized environment for engineers to execute, track, and document validation procedures.

1.2 Purpose of the Application

During commissioning, tracking hundreds of mechanical and electrical tests manually can lead to human error and disorganized records. The primary purpose of this tool is to replace scattered spreadsheets with a centralized, guided application. It ensures every piece of equipment is tested consistently and automatically generates professional deliverables for clients.

1.3 Key Features

- **Guided Test Execution:** A step-by-step wizard to record test results, attach comments, and upload photographic evidence.
- **Live Dashboards:** Real-time progress trackers at both the individual project level and the executive master level.
- **Master Specification Library:** A customizable database of reusable test cases.
- **Automated Deliverables:** Generates professional Excel workbooks and live shareable web links.

1.4 Intended Users

This tool is designed for **Commissioning Engineers** who execute the physical tests on-site, **Project Managers** who monitor overall project health, and **Validation Leads** who maintain the Master Test Case Library.

2. Login

The Login Page is the secure entry point to the application, protecting all project data and validation records

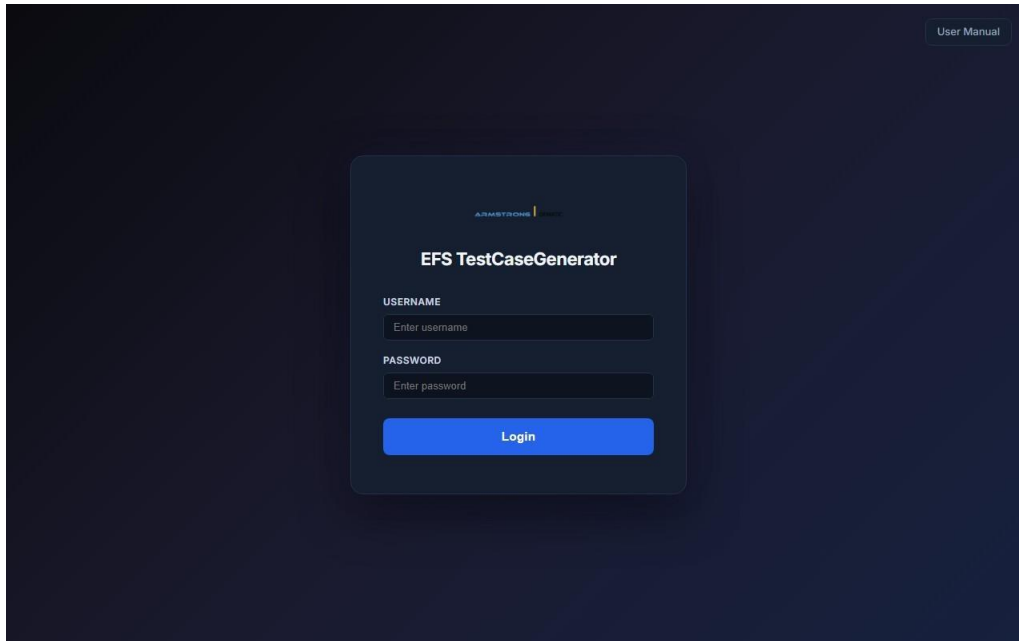


Figure 1 — Login Page Overview

2.1 Login Page Components

Username Field: Enter your designated engineering username here.

Password Field: Enter your secure password. To save time, you can press the **Enter** key on your keyboard while in this field to log in immediately.

Login Button: Click this button to authenticate your credentials and access the Project Dashboard.

User Manual Button: Located at the top right of the screen, click this button at any time to open or download this User Manual.

2.2 Common Login Tips & Troubleshooting

TIP: If you encounter an "Invalid Credentials" error, ensure your Caps Lock is turned off.

3. Project Dashboard

Immediately after logging in, you will be taken to the Project Dashboard. This serves as the central directory for all your active and historical commissioning projects.

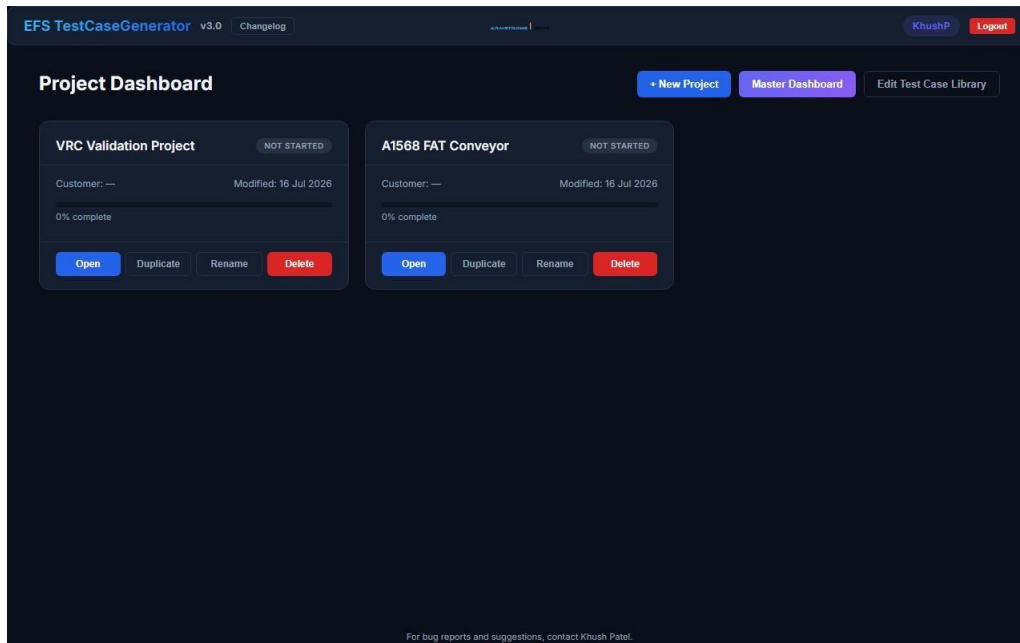


Figure 2 — Project Dashboard Interface

3.1 Header Layout

- **Version Number:** Displayed in the top left, indicating the current release of the software.
- **Changelog:** Clicking the Changelog button opens a window showing recent updates and new features added to the tool.
- **Logged-in Username:** Displayed in the top right so you can easily verify which account is currently active.
- **Logout Button:** Securely ends your current session and returns you to the Login page.

3.2 Main Controls

- **New Project Button:** Opens a prompt allowing you to create a completely blank project workspace.
- **Master Dashboard Button:** Navigates to the Executive Master Dashboard, which provides a high-level statistical overview of all projects combined.
- **Edit Test Case Library Button:** Opens the global library editor, allowing administrators to modify reusable test scopes.

3.3 Navigation & Project Cards

The main area of the dashboard is populated with **Project Cards**. Each card represents a distinct commissioning site or phase. You can scroll through the dashboard to find older projects. The cards provide quick access to open, modify, or delete a specific project workspace.

4. Project Cards

Project Cards are the individual tiles on the dashboard that represent your projects. They contain essential information and direct action buttons.



Figure 3 — Detailed View of a Project Card

4.1 Action Buttons

- **Open Project:** Enters the project workspace where you can configure zones and execute tests.
- **Duplicate Project:** Creates an exact clone of the project, including its zones and equipment selections. *Note: Execution results (Pass/Fail) are wiped clean in the duplicate, making this ideal for re-testing a site.*
- **Rename Project:** Allows you to change the title displayed on the card.
- **Delete Project:** Attempts to permanently erase the project and all its data.

4.2 Project Deletion Rules

To prevent accidental data loss, the system enforces strict deletion rules:

- **Empty Projects:** If a project has no zones and no test execution data, it can be deleted immediately upon clicking.
- **Populated Projects:** If the project contains any data, a security prompt will appear. You must manually type the master password. [This password will be given to Managers who can decide whom to forward it to]

4.3 Status Bar Behavior

The bottom of every card features a status bar showing the overall project completion percentage. This progress updates automatically as executions are completed inside the project.

- **Green Status Bar:** The bar turns solid green *only* when every single test case in the project has been executed and marked as **Passed**.
- **Yellow Status Bar:** If all test cases have been executed (100% completion) but one or more test cases have **Failed**, the bar will turn yellow.
- **Failed TC Count:** When failures exist, the card will explicitly display the count of failed test cases alongside a warning

5. Working Inside a Project

Clicking 'Open' on a project card takes you into the Project Workspace. This is where all the actual configuration and execution happens.

5.1 Project Information

At the top of the workspace, you will find the Project Information form. This captures crucial metadata about the site.

- **Project Details:** Includes fields for Customer Name, Site Location, and Project Manager.
- **Required Fields:** Some fields may be marked as mandatory. They must be filled out before you can generate the final Excel.
- **Purpose:** This data is used to automatically populate the title pages and header sections of your exported Excel workbooks and live shareable links.

5.2 Zones

A **Zone** represents a specific physical area or subsystem within the warehouse (e.g., 'Inbound Conveyors', 'Sorter 1', 'VRC A').

- **Configuration:** You create Zones using the sidebar. Each Zone acts as an independent folder containing its own set of test cases.
- **Best Practices:** Break large warehouses into logical, manageable zones. When you export to Excel, every Zone will be generated as its own separate worksheet tab.

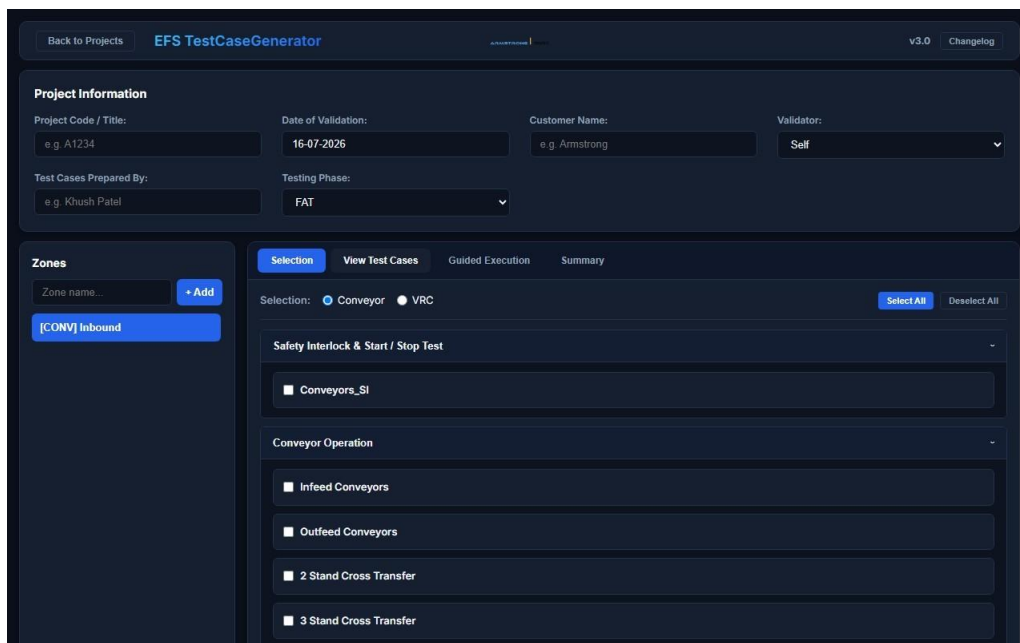


Figure 4 — Workspace Configuration - Zones and Selections

5.3 Selection Area

When you select a Zone, the main panel displays the **Selection Area**. Here, you define what equipment exists in that zone.

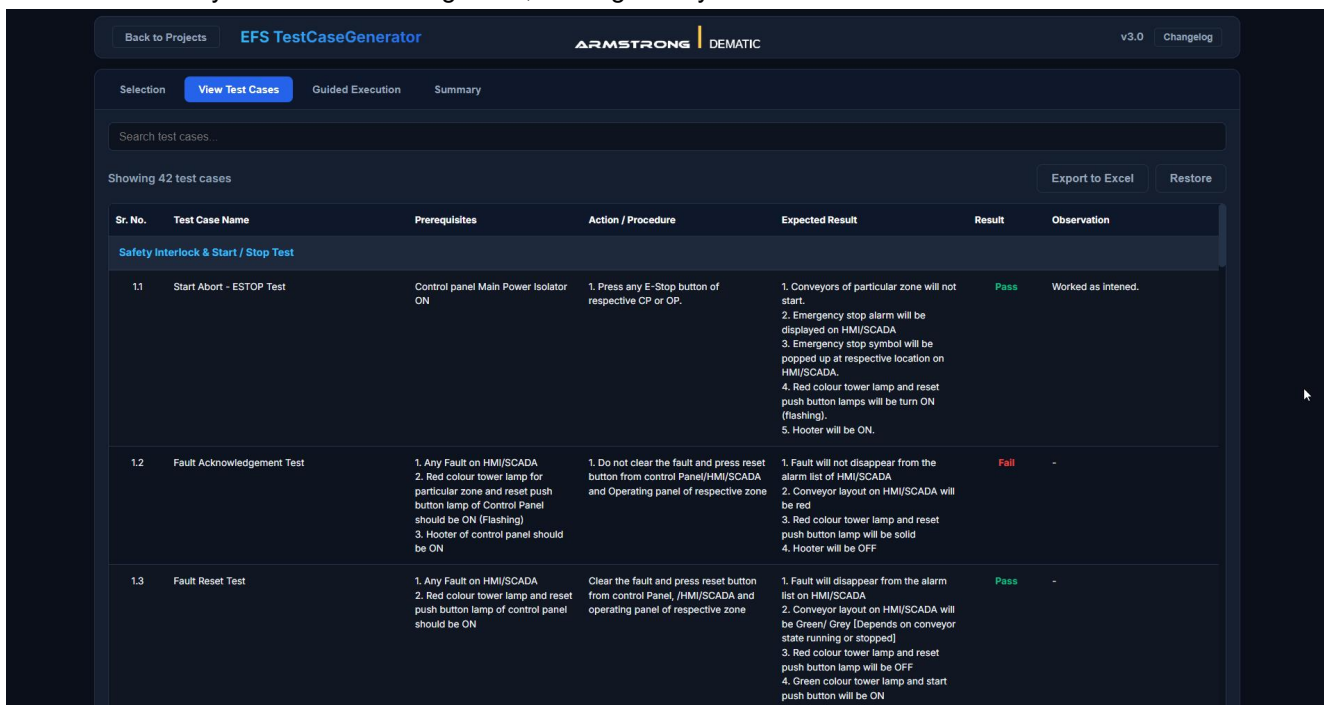
- **Selecting Clusters:** You will see checkboxes for different equipment groups (Clusters), such as 'Emergency Stops' or 'Motor Faults'. Checking a box automatically adds all associated test cases to the zone.
- **Adding/Removing Specific Tests:** You can expand a cluster and check/uncheck individual test cases to fine-tune the scope for this specific project.

IMPORTANT: Changes made in this Selection Area are **Temporary Project-Specific Changes**. Turning off a test case here only removes it from the current zone. It does not delete it from the permanent Master Test Case Library.

5.4 View Test Cases

Once your selections are made, you can review the generated list of tests.

- **Viewing & Understanding Status:** The grid displays every test case along with its current execution status (Pass, Fail, or Pending).
- **Searching & Filtering:** Use the search bar to find specific tests by keyword. Use the dropdown filters to show only 'Failed' or 'Pending' tests, making it easy to see what work remains.



Sr. No.	Test Case Name	Prerequisites	Action / Procedure	Expected Result	Result	Observation
Safety Interlock & Start / Stop Test						
1.1	Start Abort - ESTOP Test	Control panel Main Power Isolator ON	1. Press any E-Stop button of respective CP or OP.	1. Conveyors of particular zone will not start. 2. Emergency stop alarm will be displayed on HMI/SCADA 3. Emergency stop symbol will be popped up at respective location on HMI/SCADA. 4. Red colour tower lamp and reset push button lamps will be turn ON (flashing). 5. Hooter will be ON.	Pass	Worked as intended.
1.2	Fault Acknowledgement Test	1. Any Fault on HMI/SCADA 2. Red colour tower lamp for particular zone and reset push button lamp of Control Panel should be ON (Flashing) 3. Hooter of control panel should be ON	1. Do not clear the fault and press reset button from control Panel/HMI/SCADA and Operating panel of respective zone	1. Fault will not disappear from the alarm list of HMI/SCADA 2. Conveyor layout on HMI/SCADA will be red 3. Red colour tower lamp and reset push button lamp will be solid 4. Hooter will be OFF	Fail	-
1.3	Fault Reset Test	1. Any Fault on HMI/SCADA 2. Red colour tower lamp and reset push button lamp of control panel should be ON	Clear the fault and press reset button from control Panel, /HMI/SCADA and operating panel of respective zone	1. Fault will disappear from the alarm list on HMI/SCADA 2. Conveyor layout on HMI/SCADA will be Green/ Grey (Depends on conveyor state running or stopped) 3. Red colour tower lamp and reset push button lamp will be OFF 4. Green colour tower lamp and start push button will be ON	Pass	-

Figure 5 — Interactive Test Case Grid View

5.5 Guided Execution

Guided Execution is the core feature for field engineers. It provides a distraction-free, step-by-step wizard to perform the physical testing.

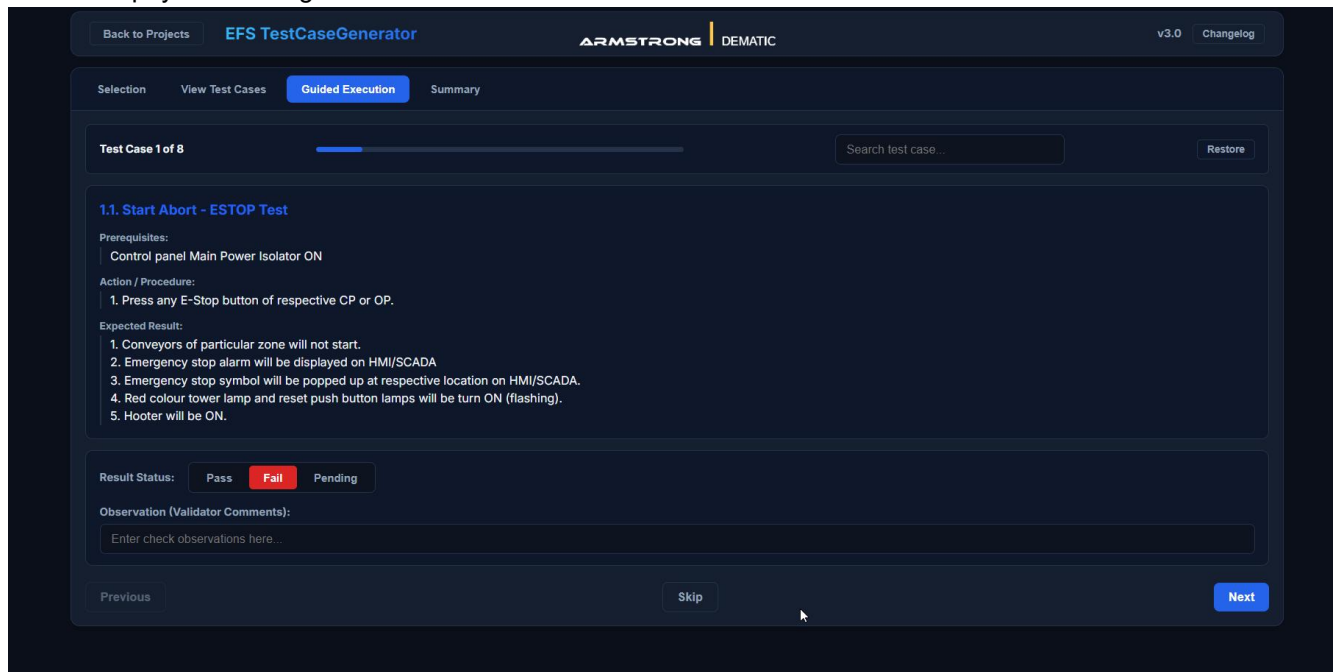


Figure 6 — Guided Execution Portal

- **How it works:** Click 'Execute' to launch the wizard. It presents one test case at a time with clear instructions and expected results.
- **Recording Pass/Fail:** Click the large green Pass or red Fail buttons. The system automatically records the result and advances to the next test.
- **Adding Comments & Evidence:** You can type technical notes into the comment box. Clicking 'Upload Evidence' allows you to attach photos of the equipment or screenshots of HMI screens.
- **Saving Progress:** Progress is saved automatically after every action. You can close the window at any time without losing data.

5.6 Summary

The Summary panel aggregates all execution data for the current project in real-time.

- **Metrics:** It displays Total Test Cases, Passed, Failed, and the overall Completion Percentage.
- **Calculation:** The statistics are calculated by scanning every individual test case across all zones. If a test case is marked Pass or Fail, it is counted as 'completed'. The completion percentage is simply the number of completed tests divided by the total number of tests.

5.7 Generate Test Cases (Deliverables)

Once testing is underway or completed, you can generate directly from the project header.

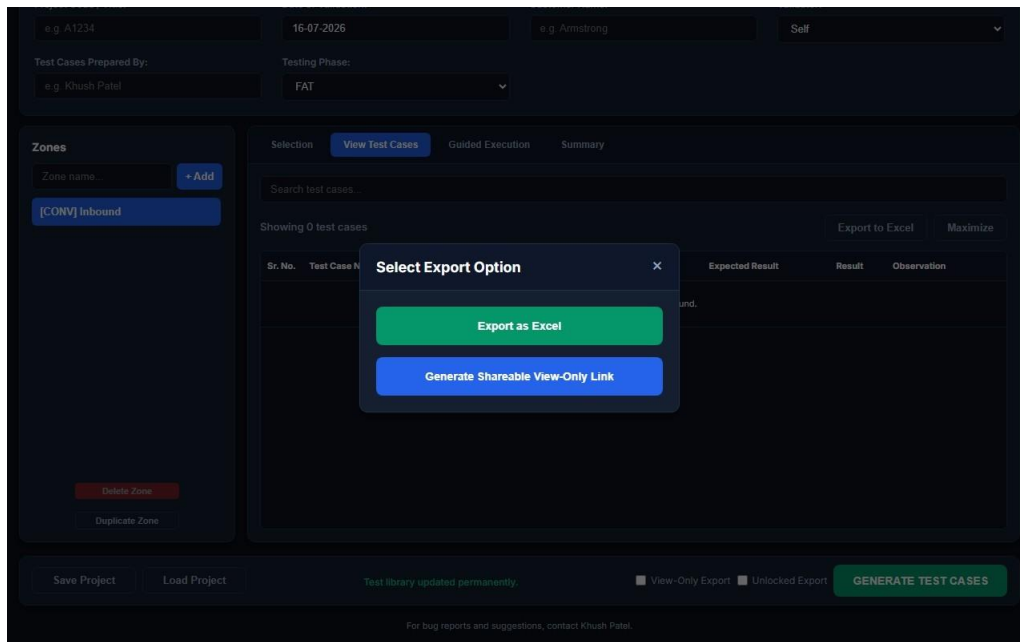


Figure 7 — Generating Deliverables

- **Generate Excel:** Creates a downloaded, formatted Excel workbook. It includes a Summary dashboard tab and individual tabs for every zone. Use this when you need an official, offline, or signed copy for the client.
- **Generate Shareable Link:** Creates a live, read-only web page. Use this to share progress with remote stakeholders. The page auto-refreshes every few seconds, allowing clients to watch the completion percentage rise in real-time.

6. Master Dashboard

The Master Dashboard is an executive summary view designed for managers monitoring multiple sites simultaneously.

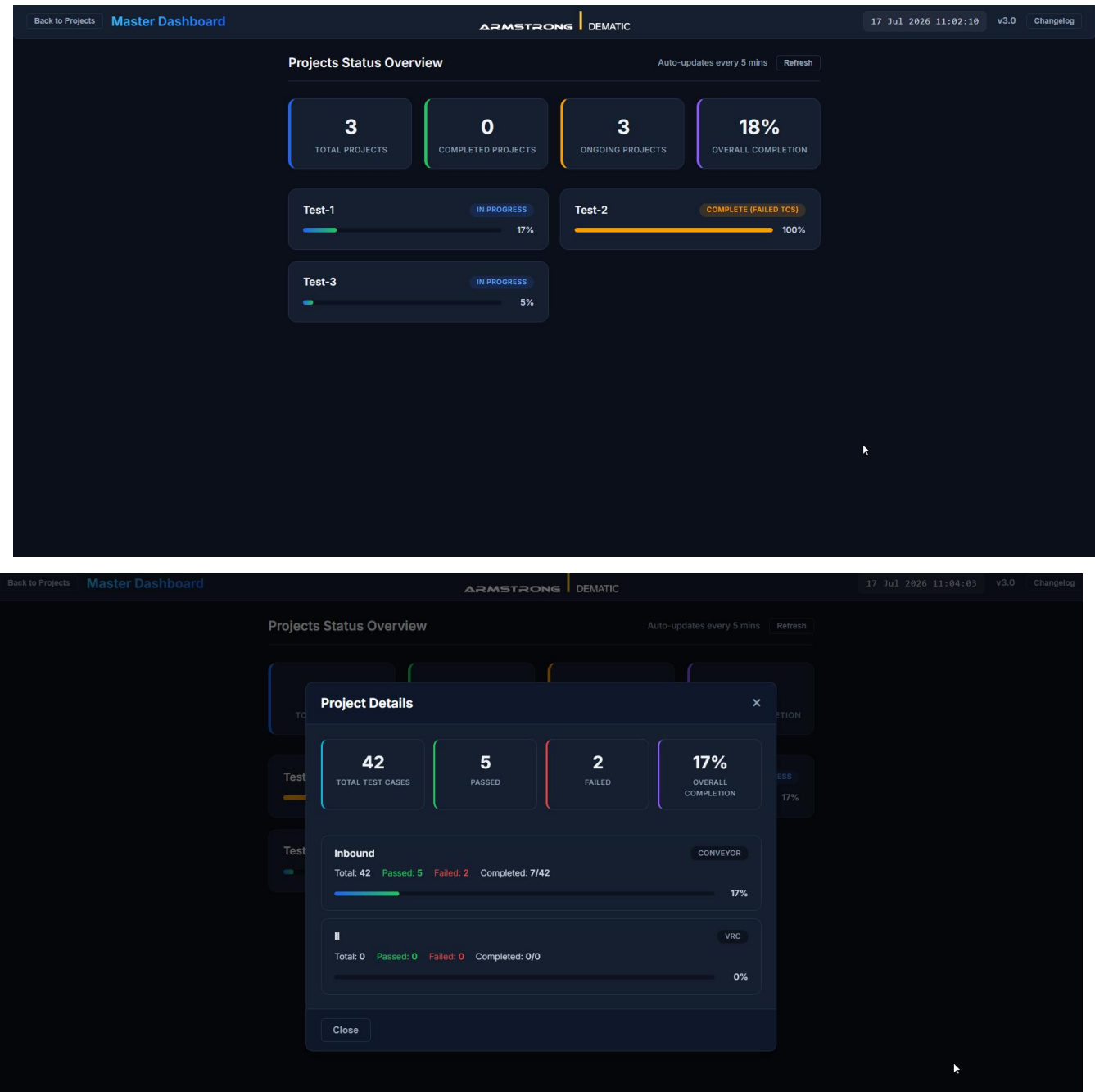


Figure 8 — Master Dashboard

- **Purpose:** To provide a bird's-eye view of all commissioning portfolios without needing to open individual projects.
- **Aggregate Metrics:** The top ribbon displays Total Projects, Ongoing Projects (in progress), Completed Projects (100% executed), and the Overall Completion Percentage across the entire company portfolio.
- **Project Progress Cards:** Below the metrics, every project is listed with its own Status bar.
- **Detailed Analytics:** Clicking on any project card opens a popup window. This provides detailed analytics for that specific project, including exact test counts and zone-by-zone progress.
- **Calculation:** Master statistics are calculated by polling the underlying execution databases of every project card dynamically every 5 minutes.

7. Edit Test Case Library

The Test Case Library is the master database of reusable test procedures. Administrators use this feature to ensure all field engineers are testing equipment against the exact same company standards.

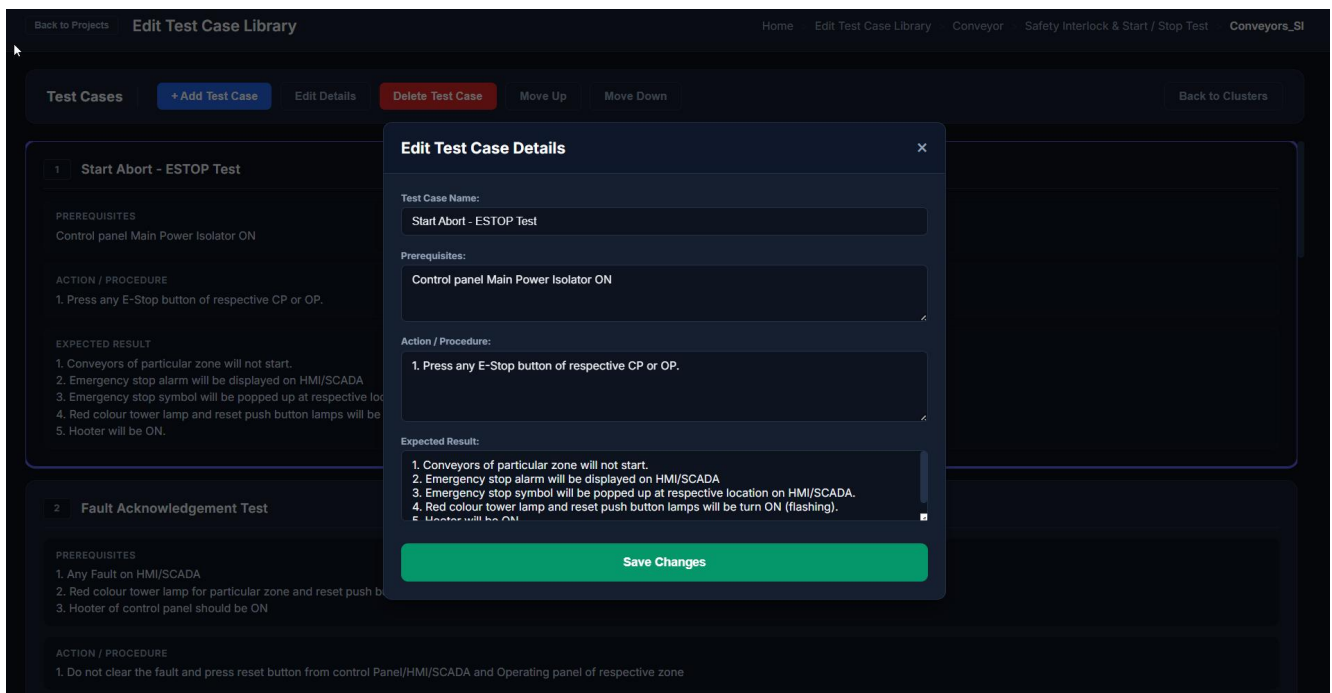
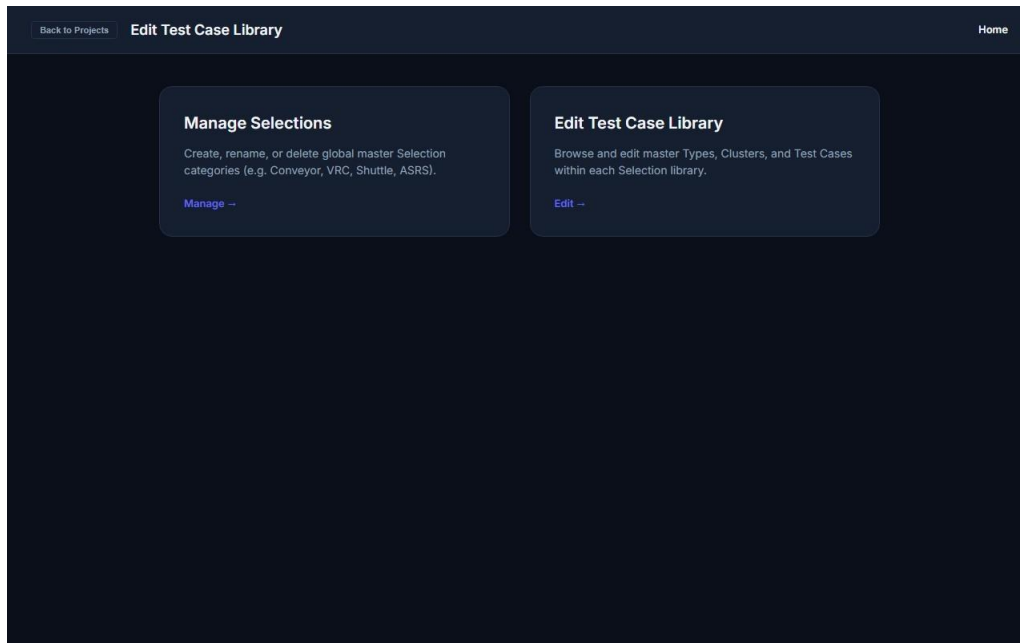


Figure 9 — Test Case Library Editor

Navigation Flow: The library is organized in a strict hierarchy:

Selections → Types → Clusters → Test Cases

7.1 Selections

Selections are the highest level categories (e.g., Conveyors vs VRCs).

- **Creating & Editing:** You can create new selections or rename existing ones.

- **Organizing Types & Clusters:** Under each Selection, you define Types (e.g., Roller vs Belt). Under Types, you manage Clusters (e.g., Safety Stops).
- **Best Practices:** Keep categories broad and logical. Too many granular Selections can make the workspace configuration confusing.

7.2 Edit Test Cases

At the bottom of the hierarchy are the actual Test Cases.

- **Creating & Editing:** You can write new test cases, define their expected results, or edit existing text for clarity.
- **Deleting:** Test cases can be permanently deleted from the library.
- **Project Modifications vs Library Modifications:** Unchecking a box in a Project Workspace only hides the test for that specific project. Deleting a test in the Edit Test Case Library removes it globally for the entire company.

WARNING: Permanent Changes: Deleting or modifying a test case in the Master Library is a permanent change. It will immediately affect all future projects that use this test case.

8. Best Practices

- **Keeping project information updated:** Fill out the Project Information form on day one to ensure deliverables are properly branded.
- **Completing guided execution regularly:** Do not wait until the end of the week to record results. Log results via Guided Execution as the tests are physically performed.
- **Reviewing Summary:** Always review the project Summary panel before closing a project to ensure no pending tests were overlooked.
- **Using the Master Dashboard:** Managers should keep the Master Dashboard open on a secondary monitor to monitor live progress.
- **Managing the Test Case Library carefully:** Because changes to the library are permanent and global, only authorized leads should make edits.
- **Backing up:** Regularly generate and save the exported Excel files to your company's secure cloud storage as a historical backup.

9. Frequently Asked Questions

Q: How do I create a project?

A: Navigate to the Project Dashboard and click the blue '+ New Project' button at the top.

Q: Why can't I delete my project?

A: If the project contains data, a security mechanism prevents accidental clicks. You must click Delete and then type the password.

Q: Why is my status bar yellow?

A: A yellow status bar indicates that 100% of the tests have been executed, but at least one test case failed.

Q: How do I duplicate a project?

A: Click the 'Duplicate' button on the project card. This copies the configuration but wipes the execution results clean.

Q: What is the difference between project test cases and the library?

A: The Library is the permanent master template. The project test cases are temporary copies generated from the library for a specific site.

Q: How do I generate Excel?

A: Inside a project, click the 'Generate Workbook' button located in the top header area.

Q: How do I use a Shareable Link?

A: Inside a project, click 'Live Report'. Copy the URL from the new browser tab and send it to your stakeholders. They do not need to log in to view it.

Q: How are completion percentages calculated?

A: Completion percentage equals the total number of executed tests (both Pass and Fail) divided by the total number of assigned tests.

10. Glossary

Selection: The highest-level equipment category in the Library (e.g., Conveyor).

Type: A sub-category within a Selection (e.g., Belt Conveyor).

Cluster: A grouping of related test cases (e.g., E-Stop Validations).

Test Case: An individual, actionable testing instruction with expected results.

Guided Execution: The step-by-step wizard interface used to record Pass/Fail results in the field.

Project Summary: The panel displaying live aggregated statistics for a single project.

Master Dashboard: The executive view displaying metrics across all active projects company-wide.

Shareable Link: A read-only, auto-refreshing web report that can be viewed without logging in.

Test Case Library: The central, permanent database storing all company testing standard

